

RESEARCH ARTICLE

Process Systems Engineering

Adaptive sampling Bayesian algorithm for constrained black-box optimization problems

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Abstract

Constrained black-box optimization (CBBO) has become increasingly popular in process optimization. Algorithms often encounter difficulties in balancing feasibility and optimality, with some even failing to find feasible solutions. This article introduces an adaptive sampling Bayesian optimization algorithm (ASBO) to solve CBBO problems effectively. The developed infill sampling criterion introduces an adaptive acquisition function to facilitate multistage optimization. The three stages consist of exploring feasible solutions, balancing feasibility and optimality, and optimizing. Furthermore, a hybrid method is proposed for complex problems. A gradient-based optimizer (GBO) aids in constructing the posterior distribution, thereby enhancing the identification of feasible regions. Additionally, four auxiliary strategies are developed to enhance stability and accelerate convergence in simulation-based optimization. The effectiveness of the proposed algorithm is validated through three benchmark problems and two process optimization cases. Comparative analysis against state-of-the-art algorithms demonstrates better iteration efficiency of ASBO algorithms.

KEYWORDS

Bayesian optimization, adaptive sampling, constrained black-box optimization, Gaussian process

1 | INTRODUCTION

Process optimization involves adjusting decision variables to achieve the minimum (maximum) objective function within predefined constraints. P1 shows a general formulation of such a problem

$$\begin{aligned} & \min_x f(x), \\ & \text{s.t.} \\ & l_m \leq g_m(x) \leq u_m, m \in \{1, 2, \dots, M\} \\ & h_k(x) = 0, k \in \{1, 2, \dots, K\} \\ & x_n^l \leq x_n \leq x_n^u, n \in \{1, 2, \dots, N\} \end{aligned} \quad (\text{P1})$$

Abbreviations: $f(x)$, Objective function; $g(x)$, Inequality constraint; $h(x)$, Equality constraint; x , Decision variables; f_i^* , Current best objective; f_{true}^* , True best objective; y , A process; $\widehat{f}(x)$, Surrogate model of object function; $\widehat{g}(x)$, Surrogate model of inequality constraint; $\widehat{h}(x)$, Surrogate model of equality constraint; Z , Maximum number of iterations; H , Parameter space; D_i , The sample set of the i th iteration; GP, Gaussian process; $AC(x)$, Acquisition function; $\mu(x)$, Mean; $\sigma^2(x)$, Variance; κ , A hyperparameter of UCB; UCB, Upper confidence bound; EI, Expective of improvement; $P_{\text{constraint}_g}$, The probability that the constraint function satisfies on $(-\infty, l)$; $P_{\text{constraint}_{gu}}$, The probability that the constraint function satisfies on $(-\infty, u)$; PFC, The difference between the upper and lower bounds; $TP_{\text{constraint}}$, The probability values PFC _{i} obtained by each constraint sum up and take the average; x_{lb} , x_{ub} , Lower and upper bounds for decision variables; x_{SL} , Decision variables obtained using GBO; $\widehat{x}_{lb}$, $\widehat{x}_{ub}$, Adjusted upper and lower bounds for decision variables.

where $f(x)$ is the objective function and x_n^l and x_n^u represent the lower and upper bounds of the variables x_n . The problem includes inequality constraints $g_m(x)$ and equality constraints $h_k(x)$. l_m and u_m represent the lower and upper bounds of the inequality constraints. In process optimization scenarios, explicit expressions of objective and constraint functions are often absent or extremely complex, thus mandating dependence on simulation models (e.g., Aspen Plus, HYSYS, and Pro/II) and algorithms for iterative refinement.¹ This category of problems, namely, constrained black-box optimization (CBBO) problems, displays features like nonconvex, multi-modality, and high evaluation costs. The problem complexity increases with the increasing n , m , and k . When $m = k = 0$, the problem is reduced to an unconstrained black-box optimization (BBO) problem.

In process optimization problems, obtaining the results of $f(x)$, $g(x)$, and $h(x)$ through black-box models generally requires high computational costs. Given the limitation of optimization algorithms in commercial process simulators, many researchers have undertaken efforts to integrate external algorithms, including both derivative-free algorithms² and gradient-based optimization (GBO) algorithms.³ Trust region algorithms, one of the GBO methods, are widely used in process optimization.^{4,5} However, these methods, relying on the finite difference approach for accurate gradient information, incur substantial computational costs and are susceptible to initial value sensitivity.⁶ Metaheuristic algorithms, such as genetic algorithm (GA), are also applied to solving BBO problems. Their iterative nature poses limitations in optimization reliant on time-consuming simulations.⁷ Christopher et al.⁸ combined the Aspen HYSYS process simulator and the particle swarm optimization (PSO) algorithm in MATLAB to optimize the propylene/propane separation configuration for the smallest TAC, reducing energy consumption by 45% and separation costs by 20%. Javaloyes et al.⁹ proposed an optimization framework based on the extended cutting plane algorithm. The proposed heuristic technologies and strategies facilitate the resolution of nonconvex simulation-based MINLP problems, and a better solution was obtained compared with the PSO algorithm and the genetic algorithm.

Surrogate-based constrained optimization (SBCO) methods effectively address complex black-box model-based design problems.¹⁰ SBCO methods replace expensive simulation models by constructing surrogate models.^{11,12} Bayesian optimization algorithm (BO) is a well-known SBCO method due to its efficacy as a global optimization algorithm. BO solves subproblems based on the surrogate model to obtain the next sampling points, subsequently using these new samples to iteratively update the surrogate model. This strategy maximizes the utilization of existing data, enabling BO to converge toward optimal solutions with fewer black-box evaluations. The corresponding cost is the iterative evaluation of sampling points through black-box simulation, with each sampling point determined by optimization.

When constraints are introduced in black-box optimization problems, the feasible region is further discretized, increasing the likelihood of getting trapped in local optimal solutions or even failing to find a feasible solution. Many methods are developed to deal with diverse constraints of CBBO problems, including penalty function, feasibility method, and ϵ -constraint method. The penalty function is

the most essential method for solving constrained optimization problems. For the Bayesian optimization algorithm, they can be divided into two categories: implicit methods and explicit methods.^{13–16} The implicit method employs an evaluation function to combine the constraint with the objective function, but the appropriate weighting between the objective and the constraint poses a challenge, which affects the quality of the results. Urm et al.¹⁷ and Ward et al.¹⁸ used the BO algorithm to solve CBBO problems by adding obstacle penalty functions to the objective function. A PET chemical recycling process and a postcombustion carbon capture system were optimized, respectively. In the feasibility-driven optimization framework developed by Tian et al.,¹⁹ the constraints were incorporated into the feasibility function to observe the degree of violation, and the overall procedure was completed through four stages. A switching strategy was developed to determine the stopping criteria for each stage. Researchers also incorporate the constraint function into the objective function to transform the problem from a single objective to a multi-objective optimization problem,^{20,21} thereby achieving a balance between minimizing the objective function and maximizing the constraint probability.

The explicit method involves solving a constraint subproblem by modeling the constraint function, a process reliant on the accuracy of the model. Based on the mean constraint criterion (MC), Priem et al.²² proposed a UTB explicit method to solve constraints, continuously adjusting the constraint learning rate (τ) during the iterative process to enhance the exploration of the feasible region. Paulson et al.²³ developed the constrained BO of computationally expensive gray-box models exploiting derivative information (COBALT) algorithm for gray-box constrained optimization problems with a probabilistic explicit constraint handling method. They proposed a modified expected improvement function and introduced a scaling factor to consider the violation probability of constraint. Lu et al.²⁴ proposed a new constrained BO strategy based on the exact penalty function and no-regret penalty algorithm. They relaxed the penalty weight factor based on the degree of constraint violation, smoothing the constraint problem to calculate the optimal solution.

Complex CBBO problems often encounter difficulties where initial random samples fail to yield feasible solutions, potentially causing algorithms reliant on feasible solutions to fail. To address this challenge, researchers have proposed several strategies, such as strategically adding sampling points in areas where feasibility probability is maximized and designing a merit function to identify local feasible areas and explore promising areas.^{25,26} In a constrained adversarially robust Bayesian optimization (CARBO) developed by Kudva et al.,²⁷ the objective function is initially ignored to maximize the probability of robust constraint satisfaction. Another approach involves step-by-step optimization, initially addressing constraint violations before improving the objective function within the feasible region,²⁸ while efficient algorithms and advanced BO methods proposed by scholars effectively address many benchmark problems.^{29,30} However, for complex problems, such as process optimization problems, the feasibility and the efficiency of the algorithms may not be guaranteed. The algorithm performances are impacted by the simulators, where simulation failures may arise. In addition, the feasible region of the constrained optimization can be very limited compared to

the variable space, which leads to a significant number of iterations of Bayesian optimization. Consequently, it is necessary to develop an efficient and robust algorithm capable of converging to a local optimum or the global optimum in fewer iterations.

We propose an adaptive sampling Bayesian optimization algorithm (ASBO) for solving CBBO problems. Firstly, an efficient infill sampling criterion is introduced, combining the upper confidence bound (UCB) and probability of feasibility (PF) methods. The sampling strategy enables a multistage optimization, progressing through stages that prioritize exploring feasible solutions, balancing feasibility with optimality, and finally optimizing the objective. To mitigate cold-start challenges, we develop a hybrid algorithm in which a gradient-based optimizer generates improved initial samples to enrich the Bayesian algorithm's posterior distribution. Additionally, four auxiliary strategies are constructed to enhance stability and expedite convergence in simulation-based optimization, including initial point generation for subproblems, a domain restart strategy, equality constraint relaxation, and parallel computing. The effectiveness of the algorithms is demonstrated by three benchmark problems and two chemical industry applications. A complex large-scale solution polyethylene process optimization problem validates the advantages of the proposed hybrid method.

2 | PRELIMINARY

2.1 | BO Algorithm

BO is an effective global optimization algorithm.¹³ Algorithm 1 shows the pseudocode of BO for the problem defined in P1. The surrogate models $\widetilde{f}(\mathbf{x})$, $\widetilde{g}(\mathbf{x})$, and $\widetilde{h}(\mathbf{x})$ are constructed through the Gaussian processes (GPs), which approximate the objective function $f(\mathbf{x})$, equality constraints $g(\mathbf{x})$, and inequality constraints $h(\mathbf{x})$, respectively. In the initial stage, a random sampling method (e.g., Latin hypercube sampling LHS) is used to select L initial sample points $\mathbf{x}_0$ from the decision variable space H . These initial samples form the set D_0 , which is used to update the surrogate models. In the following iteration process, the next sampling point $\mathbf{x}_i$ is obtained by solving a subproblem that minimizes the acquisition function based on the current surrogate models. The black-box model is then carried out to evaluate $f(\mathbf{x}_i)$, $g(\mathbf{x}_i)$, and $h(\mathbf{x}_i)$, with these results added to the sample set D_i . The surrogate models are iteratively refined with each updated sample set D_i , and this process repeats until the maximum number of iterations Z is reached. The feasible solution with the minimum objective value is returned as the final output.

A Gaussian process³¹ (GP) is determined by the mean function $\mu(\mathbf{x})$ and the kernel function $k(\mathbf{x}, \mathbf{x}')$ (covariance function) of a real process y . In Algorithm 1, the objective function $f(\mathbf{x})$ and the constraint functions $g(\mathbf{x})$ and $h(\mathbf{x})$ are modeled as GPs individually. Taking the objective function as an example, the GP is defined as follows:

$$y = f(\mathbf{x}) \sim \text{GP}(\mu(\mathbf{x}), k(\mathbf{x}, \mathbf{x}')), \quad (1)$$

where $\mu(\mathbf{x}) = E[f(\mathbf{x})]$, $k(\mathbf{x}, \mathbf{x}') = E[(f(\mathbf{x}) - \mu(\mathbf{x}))(f(\mathbf{x}') - \mu(\mathbf{x}'))]$. The Gaussian process uses maximum likelihood estimation to obtain the optimal

Algorithm 1

Input: Objective function $f(\mathbf{x})$ and constraint functions $g(\mathbf{x})$ and $h(\mathbf{x})$; number of initial samples L ; maximum number of iterations Z ; and initial surrogate models $\widetilde{f}(\mathbf{x})$, $\widetilde{g}(\mathbf{x})$, and $\widetilde{h}(\mathbf{x})$ using GPs.

for $i = 0, 1, 2, 3 \dots Z$ **do**.

1: **If** $i = 0$

Randomly select initial sample points $\mathbf{x}_0$ from H .

Else:

Find $\mathbf{x}_i$ by solving a subproblem: minimizing the acquisition function.

2: Evaluate the black-box functions $f(\mathbf{x})$, $g(\mathbf{x})$, $h(\mathbf{x})$ at $\mathbf{x}_i$.

3: **If** $i = 0$:

Initialize $D_i = \{\mathbf{x}_i \sim f(\mathbf{x}_i), g(\mathbf{x}_i), h(\mathbf{x}_i)\}$

Else:

Augment the dataset with the new observation:

$D_i = D_{i-1} \cup \{\mathbf{x}_i \sim f(\mathbf{x}_i), g(\mathbf{x}_i), h(\mathbf{x}_i)\}$.

4: Update GP models $\widetilde{f}(\mathbf{x})$, $\widetilde{g}(\mathbf{x})$, $\widetilde{h}(\mathbf{x})$ based on D_i .

5: $i = i + 1$

end for

output: $\mathbf{x}^*$, $f(\mathbf{x}^*)$ is the minimum feasible solution found in D_Z .

hyperparameters of the kernel function. In the sample set with L evaluated data points $D_L = \{\mathbf{x}_k, y_k\}_{k=1, \dots, L}$, the posterior probability distribution $p(y|\mathbf{x}, D_L)$ of y is shown in the following equation:

$$p(y|\mathbf{x}, D_L) = \mathcal{N}(y|\mu(\mathbf{x}), k(\mathbf{x}, \mathbf{x}')). \quad (2)$$

Each point in the design space of a Gaussian process is associated with a normally distributed random variable. The joint distribution of all these random variables across the space constitutes a multivariate Gaussian distribution. The joint Gaussian distribution of the known sample set D_L and the new sampling point $(\mathbf{x}_{L+1}, y_{L+1})$ satisfies Equation (3) and adheres to the independent and identically distributed condition. Here, $\mathbf{x}_{1:L}$ and $y_{1:L}$ represent the evaluated sampling points and their corresponding function values, $\mathbf{x}_{L+1}$ and y_{L+1} , represent the next sampling point and its predicted function value. In the covariance matrix of Equation (3), $K = k(\mathbf{x}, \mathbf{x})$, $K^* = [k(\mathbf{x}_{L+1}, \mathbf{x}_1), k(\mathbf{x}_{L+1}, \mathbf{x}_2) \dots k(\mathbf{x}_{L+1}, \mathbf{x}_L)]$, $K^{**} = k(\mathbf{x}_{L+1}, \mathbf{x}_{L+1})$

$$\begin{bmatrix} y_{1:L} \\ y_{L+1} \end{bmatrix} \sim \mathcal{N}\left(\begin{pmatrix} \mu(\mathbf{x}_{1:L}) \\ \mu(\mathbf{x}_{L+1}) \end{pmatrix}, \begin{pmatrix} K & K^* \\ K^{*T} & K^{**} \end{pmatrix}\right). \quad (3)$$

Different from the deterministic model, the output of GPs at a given point is represented by the mean $\mu(\mathbf{x}_{L+1}) = K^{*T} \begin{pmatrix} K & K^* \\ K^{*T} & K^{**} \end{pmatrix}^{-1} y_{1:L}$

and the variance $\sigma^2(x_{L+1}) = K_{**} - K_*^T \begin{pmatrix} K & K_* \\ K_*^T & K_{**} \end{pmatrix}^{-1} K_*$. These are calculated from the joint distribution. The mean μ represents the predicted value of the model for a given input and serves as the conditional expectation of the target variable. The variance σ^2 represents the uncertainty of prediction for a given input.

2.2 | Subproblem: minimizing acquisition function

The optimization of the subproblem in Algorithm 1 is a crucial step in BO,³² which determines the next sampling points. We employ the minimization of the acquisition function $AC(x)$ as the subproblem, as shown in the following equation:

$$x_i = \arg \min_x AC(x). \quad (4)$$

A well-designed acquisition function balances the trade-off between “exploitation” and “exploration” processes. A deterministic algorithm is utilized to achieve optimal results, with the decision variable x_i representing the next sampling point.

Common methods for constructing surrogate models include upper confidence bound (UCB),³³ expected improvement (EI),³⁴ and probability of improvement (PI).³⁵ The EI criterion addresses a limitation of the PI criterion, which does not consider the degree of improvement. EI recommends the next sampling point based on the expectation of improvement, as shown in Equation (5), where $\Phi()$ is the cumulative distribution function (CDF), $\phi()$ is the probability density function (PDF) of the standard normal distribution, and f_i^* is the current best objective:

$$EI = \begin{cases} (f_i^* - \mu(x)) \Phi\left(\frac{f_i^* - \mu(x)}{\sigma(x)}\right) + \sigma(x) \phi\left(\frac{f_i^* - \mu(x)}{\sigma(x)}\right) & \text{if } \sigma > 0 \\ 0 & \text{if } \sigma = 0 \end{cases}. \quad (5)$$

The UCB criterion determines the next sampling point by setting a hyperparameter κ to balance the mean $\mu(x)$ and the standard deviation $\sigma(x)$. The function is expressed as follows:

$$UCB = \mu(x) + \kappa \cdot \sigma(x). \quad (6)$$

Both the EI and UCB functions were proposed based on unconstrained problems, where only the objective function is considered in the acquisition function. In constrained optimization, however, the acquisition function should consider both objective improvement and feasible region exploration.³⁶ A common approach combines the objective function with the degree of constraint violation. The probability of feasibility (PF) method³⁷ calculates the probability that the constraints are satisfied under the current decision variables. Taking $l \leq g(x) \leq u$ as an example, it calculates the mean μ and the standard deviation σ at $x \in D$ through the Gaussian model of the constraint function $\tilde{g}(x)$ and constructs a cumulative distribution function to estimate the probability that the

constraint function is satisfied within the intervals $(-\infty, l)$ and $(-\infty, u)$. The difference between the upper and lower bounds yields the probability of satisfying the constraint at the decision variable x , namely, PFC , as shown in the following equation:

$$\begin{aligned} P_{\text{constraint}_{lb}} &= \text{CDF}(lb, \mu, \sigma) \\ P_{\text{constraint}_{ub}} &= \text{CDF}(ub, \mu, \sigma) \\ PFC &= P_{\text{constraint}_{ub}} - P_{\text{constraint}_{lb}} \end{aligned} \quad (7)$$

The PF method commonly multiplies the probability of each constraint to obtain the total probability value of satisfying all constraints. However, in cases with multiple constraint functions, the multiplication can drive the PF close to 0,³⁸ especially when the probability of any one constraint is extremely low. We define the total probability $TP_{\text{constraint}}$ as the average of PFC_i obtained by all constraints, as shown in Equation (8), ensuring that the overall probability is not disproportionately influenced by any single constraint

$$TP_{\text{constraint}} = \frac{1}{n} \sum_i^n PFC_i. \quad (8)$$

Strategies of combining the PF method with the EI function have been developed to handle constraints,^{39,40} which can achieve a balance between minimizing the objective function and maximizing the constraint probability. However, the result of the EI function is the degree of the objective function improvement. The result of the UCB function is close to the true value of the objective function, which is conducive to the observations. The hyperparameters of the acquisition function combining the UCB function with the PF method are easier to determine than those for the EI function and the PF method. The acquisition function can be configured to explore the feasibility, balance feasibility, and optimality or optimize the objective.

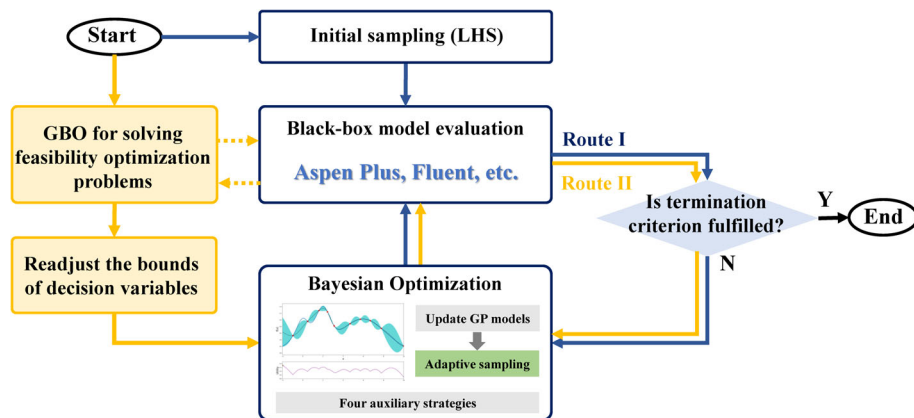
3 | METHODOLOGY

The proposed method integrates multiple effective strategies to achieve an efficient and robust algorithm. Figure 1 illustrates the proposed ASBO algorithm, which includes two routes, namely, ASBO-I and ASBO-II.

3.1 | Multistage black-box optimization

In Route I (blue solid line), LHS is used to randomly generate initial sample points and form the data set D_0 , which is used for constructing the initial posterior distribution. The ASBO algorithm then recommends new sampling points iteratively until the termination criterion is met. In the proposed algorithm, the iteration halts after Z iterations. The ASBO-I algorithm follows the same procedure as Algorithm 1 introduced above, with improvements made through an adaptive acquisition function and several auxiliary strategies (as shown in Sections 3.2 and 3.3). Route I can successfully solve most CBBO problems. However, different from gradient-based methods, surrogate-based methods

FIGURE 1 ASBO-I (Route I) and ASBO-II (Route II) for solving CBBO problems.



encounter challenges of high cold-start costs and are significantly affected by parameter space. The so-called cold-start is due to the lack of effective information in the initial stage, leading to a substantial disparity between the surrogate model and the actual model. In highly complex problems, the initial posterior distribution generated through random sampling often provides inadequate guidance for the subsequent process, leading to convergence failure.⁴¹ Therefore, enhancing the accuracy of the posterior distribution can greatly improve the exploration of the feasible region. A demo case is shown in Section S1 of the Supporting Information.

When ASBO-I fails to find a feasible solution after Z iterations, Route II (yellow solid line), referred to as ASBO-II, is activated. Feasibility problems derived from the initial problems are solved by a GBO algorithm (e.g., SLSQP). The reason for solving feasibility problems is that GBO algorithms typically compute the gradients of the objective and constraint functions using the finite difference method, gradually approaching the feasible region. However, if the gradient descent direction of the constraint function conflicts with the objective function, the algorithm may prematurely terminate. By setting the objective function to a constant, only the gradient direction of the constraint is considered, significantly simplifying the problem. Even in cases of nonconvergence, the result can still be utilized as the initial sample set. In Route II, the feasibility problem is solved K times by the GBO algorithm with random points as the initial values. The decision variable bounds are then adjusted based on the results of the feasibility problems. Section-S2 of the Supporting Information shows the pseudocode for the specified rules. Compared to random sampling points, the results from the feasibility problems are closer to the feasible region, offering more effective information for the posterior distribution. Warm starts and adjusting the decision variable bounds can improve the efficiency of the proposed BO algorithm. ASBO-I and ASBO-II are summarized in Algorithm 2. Notably, ASBO-II can be directly employed, as shown in Figure 1.

3.2 | Adaptive acquisition function

In each iteration process of the BO algorithm, the next sampling point is determined by optimizing the acquisition function as shown in

Algorithm 2 The ASBO algorithm

1. Start ASBO-I with the variable bounds x_{lb} and x_{ub} and carry out Z iterations using Algorithm 1 introduced above.
2. If feasible solution is obtained after Z iterations:
Output: $x^*, f(x^*)$ from D_Z , $f(x^*)$ is the feasible solution with the minimum objective function.
3. Else if no feasible solution is obtained after Z iterations, ASBO-II will be activated to solve the CBBO problem
 - (1) Solve the feasibility problem using GBO by K times.
 - (2) Obtain K sets of solutions x_{SL} and dataset D_{SL} .
 - (3) Adjust the decision variable bounds: $\hat{x}_{lb}, \hat{x}_{ub}$.
 - (4) Restart ASBO-I for another Z' iteration with L initial samples, obtain dataset $D_{Z'}$.
 - (5) **Output:** $x^*, f(x^*)$ from D_{SL} and $D_{Z'}$, $f(x^*)$ is the one feasible solution with the minimum objective function.

Equation (9), where the acquisition function is calculated using the surrogate models $\widetilde{f}(\mathbf{x})$, $\widetilde{g}(\mathbf{x})$, and $\widetilde{h}(\mathbf{x})$ of the objective and constrain functions. A three-stage acquisition function is proposed to balance the improvement of the objective and constraint functions and accelerate the exploration of the feasible region. We design the acquisition function based on the UCB + PF criterion, as shown in Equation (10):

$$x_i = \arg \min_{\mathbf{x}} AC(\widetilde{f}(\mathbf{x}), \widetilde{g}(\mathbf{x}), \widetilde{h}(\mathbf{x})), \quad (9)$$

$$AC = \begin{cases} \text{UCB}(\mu, \sigma, \kappa) + \frac{1}{s + TP_{\text{constraint}}^{m_1}} & TP_{\text{constraint}} < P_1 \\ \text{UCB}(\mu, \sigma, \kappa) + \frac{1}{TP_{\text{constraint}}^{m_2}} & P_1 < TP_{\text{constraint}} < P_2 \\ \text{UCB}(\mu, \sigma, \kappa) & P_2 < TP_{\text{constraint}} \end{cases}$$

$$m_1 = 5, m_2 = 2, s = 10^{-5}, \kappa = 2.5, P_1 = 0.78, P_2 = 0.95. \quad (10)$$

The input of the acquisition function includes $\text{UCB}(\mu, \sigma, \kappa)$, $TP_{\text{constraint}}$, and six hyperparameters (m_1 , m_2 , s , κ , P_1 , and P_2). The mean

μ and the standard deviation σ are calculated by the surrogate model $\widehat{f}(x)$ of the objective function, while the $TP_{\text{constraint}}$ can be obtained by Equation (8) using the surrogate model $\widehat{g}(x)$ and $\widehat{h}(x)$. The hyperparameter s is used to prevent a tremendous value when $TP_{\text{constraint}}$ is small. The lower and upper probability thresholds P_1 and P_2 are used to discriminate variable stages. The hyperparameters m_1 and m_2 are used to adjust the influence of feasibility in various stages. Common hyperparameters are also given in Equation (10). The sensitivity of the hyperparameters and the determination of hyperparameters can be found in the case study section. The process of minimizing the acquisition function is divided into three stages, guided by the following principles.

3.2.1 | Exploring feasible solutions

When the probability of satisfying the constraints ($TP_{\text{constraint}}$) is less than the predefined lower threshold P_1 , an increase in $TP_{\text{constraint}}$ has a significant impact on the acquisition function, thereby facilitating the discovery of feasible solutions.

3.2.2 | Balancing feasibility and optimality

Once $P_{\text{constraint}}$ exceeds P_1 , indicating a high likelihood of satisfying the constraints, the effect of increasing $TP_{\text{constraint}}$ on the acquisition function becomes smaller than in the first stage, facilitating a balance between enhancing the feasibility and optimizing the objective function.

3.2.3 | Optimizing objective function

Once $P_{\text{constraint}}$ exceeds the upper threshold (P_2), indicating a high probability of meeting the constraints, the focus shifts to minimizing the objective function as much as possible.

3.3 | Auxiliary strategy

3.3.1 | Initial point generation for subproblem

Minimizing the acquisition function (Equation (9)) is a nonconvex non-linear problem, making it challenging to obtain a globally optimal solution. Deterministic algorithms typically solve Equation (9) with multiple initializations. We use three types of initial points to improve the efficiency and results. The first type involves randomly selecting six decision variables to ensure global exploration. The second type uses the LHS method to sample 10,000 data points, evaluates the acquisition function value, and then selects the six sample points with the lowest values. The third type selects up to six solutions from the evaluated dataset, which have objective values better than the current best result and a relative error of no more than 1% between the

constraint calculation value and the constraint boundary value. This method balances the modes of “exploration” and “utilization”, selecting higher-quality initial values for iteration, while also revisiting promising results for further exploration. The comparison results of whether to use the optimization strategy and different algorithms are shown in Section S3 of the Supporting Information.

3.3.2 | Neighborhood restart strategy

For CBBO problems involving process simulators, process convergence failure cannot be ignored. Common methods to address this issue include returning a set of obstacle results or restarting using previous results,^{9,42} leading to deviations and reduced efficiency. The ASBO algorithm introduces a domain restart strategy, where unconverged results are discarded, and the simulation is restarted in the vicinity of the current decision variable. The maximum restart number is set to 10. A poor result will be chosen as the obstacle function value if no convergence result is obtained. It forms a barrier in the current variable space to avoid wasting computational resources in the following optimization. However, since restarting simulation generally requires higher computational costs, this strategy can be enabled on demand.

3.3.3 | Constraint relaxation

Rounding errors may arise due to the numerical accuracy limitations of the algorithm, leading to small deviations in the numerical results. These errors could cause potentially promising solutions to be overlooked. To address this, the constraint relaxation method is employed, considering solutions with deviations of less than 1 % from the constraint lower or upper bounds as feasible. This strategy is specifically applied to black-box models in simulation software to mitigate the impact of numerical accuracy from data exchange. Additionally, based on the characteristics of the PF criterion, equality constraints are relaxed into inequality constraints, where deviations of less than 1% from equality constraints are considered feasible.

3.3.4 | Parallel computing

The proposed acquisition function involves multiple hyperparameters. Different sampling points can be obtained by solving subproblems with various hyperparameters. Instead of fine-tuning hyperparameters sequentially, multiple sampling points can be obtained by acquisition functions with different hyperparameters. These points can then be evaluated simultaneously by the black-box models through parallel computing. Using multi-core for parallel computing on a single computer is an effective way to improve the computing speed.⁴³ In BO, parallel computing necessitates waiting for all ongoing processes to be completed before updating the posterior distribution. While parallel computing does not fully leverage the results of each call to the

black-box model compared to single-point computation, inputting batch data points into the black-box model can still improve the optimization efficiency.^{44,45} For example, running four different Aspen Plus simulations in parallel using four CPU cores is faster than running the same four simulations sequentially on the same cores.

4 | CASE STUDIES

The effectiveness and advantages of the proposed algorithm are demonstrated through five cases. Cases 1–3 are explicit mathematical benchmark problems, which are treated as black-box functions in this study. Cases 4 and 5 are process optimization problems using Aspen Plus as process simulators. Case 4 is an extractive distillation unit problem from Ma et al.,⁴⁶ while Case 5 is a solution polymerization process optimization problem from the industry. ASBO-I algorithm is applied to solve Cases 1–4, while ASBO-II is used for Case 5.

4.1 | Cases 1–3 (benchmark problems)

Cases 1 and 2 are well-known problems whose feasible regions and global optimal solutions are known. Case 3 is a newly constructed problem based on a few widely used functions. Case 1 is a nonlinear, nonconvex two-dimensional problem (NNP),⁴⁷ which contains three inequality constraints. The global optimal solution is: $y_{\text{global}} = 0, x_{\text{global}} = [-4, -6]$. Due to the inherent numerical differentiation problem of the algorithm, the global optimal solution is considered to be found when the objective function value y is less than 10^{-5} .

$$\begin{aligned} \min y &= (x_1 + 4)^2 + (x_2 + 6)^2 \\ \text{s.t. } &-2x_1 + x_2 - 15 \leq 0 \\ &\frac{x_1^2}{2} + 4x_1 - x_2 - 5 \leq 0 \\ &\frac{(x_1 - 4)^2}{5} - 2x_2^2 + 10 \leq 0 \\ &-10 \leq x_1 \leq 5 \\ &-15 \leq x_2 \leq 15 \end{aligned} \quad (\text{Case 1})$$

Case 2 shows the new Branin problem,³⁹ where the constraint function is modified from the traditional Branin function. The global optimal solution is $y_{\text{global}} = -268.78, x_{\text{global}} = [3.273, 0.0489]$:

$$\begin{aligned} \min y &= -(x_1 - 10)^2 - (x_2 - 15)^2 \\ \text{s.t. } &\left(x_2 - \frac{5.1}{4\pi^2}x_1^2 + \frac{5}{\pi}x_1 - 6\right)^2 + 10\left(1 - \frac{1}{8\pi}\right)\cos x_1 + 5 \leq 0 \\ &-5 \leq x_2 \leq 10 \\ &0 \leq x_1 \leq 15 \end{aligned} \quad (\text{Case 2})$$

Case 3 presents a new hybrid Ackley–Branin problem (NHAB). This problem includes four constraint functions, namely, C1, C2, C3, and C4. The objective and constraint function C1 are modified from the Ackley and Branin functions, respectively, with both functions

formulated five dimensions. The NHAB problem introduces greater complexity compared to Cases 1 and 2, due to its higher-dimensional objective function and the inclusion of multiple intricate constraints:

$$\begin{aligned} \min f_{\text{Ackley}}(x) &= -20 \times \exp\left(-0.2 \times \sqrt{\frac{\sum_{i=1}^5 x_i^2}{5}}\right) - \exp\left(\frac{\sum_{i=1}^5 \cos(2\pi x_i)}{5}\right) \\ &\quad + 20 + \exp(1) \\ \text{s.t. } C_1(x) &= \left(x_2 - \frac{5.1}{4\pi^2}x_1^2 + \frac{5}{\pi}x_1 - 6\right)^2 + 10\left(1 - \frac{1}{8\pi}\right)\cos(x_1) + \\ &\quad \left(x_5 - \frac{5.1}{4\pi^2}x_4^2 + \frac{5}{\pi}x_4 - 6\right)^2 + 10\left(1 - \frac{1}{8\pi}\right)\cos(x_4) - 80 \\ C_2(x) &= \sum_{i=1}^5 \sin(2\pi x_i) - \prod_{i=1}^5 \cos(3\pi x_i) \\ C_3(x) &= \sum_{i=1}^5 \exp(-x_i^2) + \prod_{i=1}^5 \sin(x_i) \\ C_4(x) &= \frac{1}{2}\sin(2\pi(x_1 - 2x_2)) + x_1 + 2x_2 - 1.5 + 3x_3 - 2x_4 + 0.4x_5^2 \\ &\quad -5 \leq x_1 \leq 12 \quad -1 \leq C_1 \leq 2 \\ &\quad -20 \leq x_2 \leq 22 \quad -1 \leq C_2 \leq 4 \\ x &= [x_1, x_2, x_3, x_4, x_5] \quad -10 \leq x_3 \leq 10 \quad -2 \leq C_3 \leq 3 \\ &\quad -12 \leq x_4 \leq 20 \quad 0 \leq C_4 \leq 1.5 \\ &\quad -15 \leq x_5 \leq 18 \end{aligned} \quad (\text{Case 3})$$

4.1.1 | Comparisons of ASBO, CBO, SCBO, and FDBO

The performance of ASBO in Cases 1–3 is compared with three state-of-the-art BO algorithms reported in the literature: CBO,⁴⁸ SCBO,⁴⁹ and FDBO,¹⁹ all of which are capable of handling inequality constraints. For four BO algorithms (ASBO-I, CBO, FDBO, and SCBO), the number of initial points (50, 100, and 100 for Cases 1–3, respectively) and the maximum iterations (50, 50, and 200 for Cases 1–3, respectively) are kept consistent across all algorithms. The iterations of ASBO-I, CBO, FDBO, and SCBO correspond to the number of black-box function evaluations, which is an important metric for complex black-box models. The performance of these four BO algorithms is compared by the base-10 logarithm of the best-sample simple regret,²³ which is defined as follows:

$$\log_{10} \text{Regret}_i = \log_{10}(f_i^* - f_{\text{ture}}^*), \quad (11)$$

where f_i^* and f_{ture}^* represent the current best objective in the i th iteration and the true best objective. Since BO algorithms depend on randomly selected initial points, every algorithm is repeated 100 times to estimate the average $\log_{10} \text{Regret}_i$ for each algorithm. Error bars are computed by estimating the confidence intervals as 1.96 times the standard deviation divided by the square root of the number of repeats. Note that failed runs are excluded when calculating the standard deviation. Figure 2 shows the regret plot of four BO algorithms,

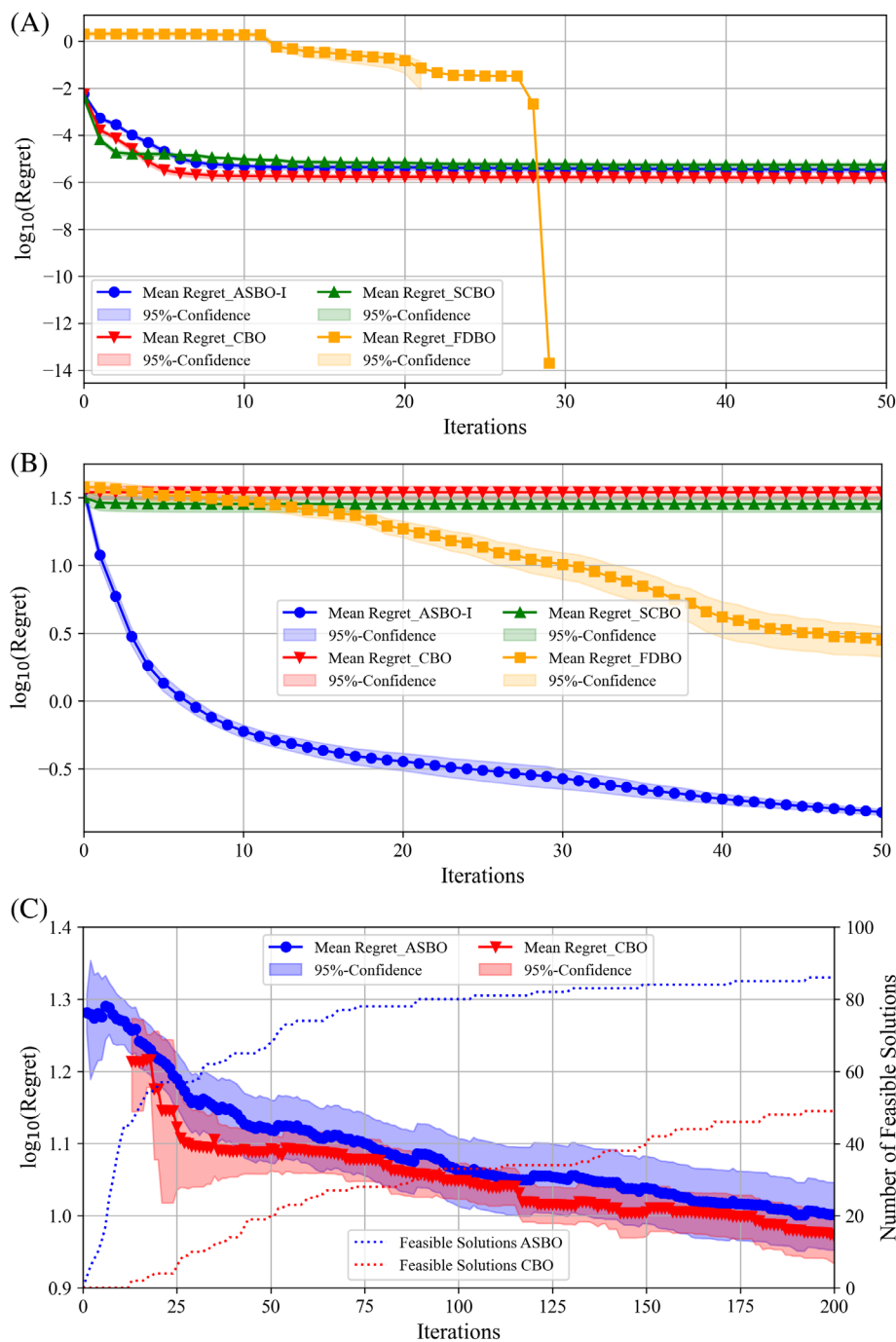


FIGURE 2 The $\log_{10}(\text{Regret})$ via ASBO-I, CBO, SCBO, and FDBO in Case 1 (A), Case 2 (B), and Case 3 (C).

where ASBO-I, CBO, SCBO, and FDBO are represented by blue, red, green, and yellow, respectively. The solution results are shown in Table 1, including the average iterations and average solutions.

For the NNP problem (Case 1), most algorithms obtain the global optimal solution, as shown in Table 1. The regret plot in Figure 2A demonstrates that ASBO-I, CBO, and SCBO perform similarly for this relatively simple case, reaching solutions close to optimal solutions within less iterations compared to FDBO. It can be observed that the results of the first 10 iterations by FDBO are relatively worse than those of the other three algorithms, due to its stage-based nature of FDBO. In 100 independent runs, FDBO reaches its convergence criterion and halts before the maximum number of iterations. The

constraint functions of the new Branin problem (Case 2) are more complex than the one of Case 1, highlighting the performance differences of each algorithm, as shown in Figure 2B. The results of ASBO-I are closest to the global optimum. The regret plot shows CBO and SCBO quickly converge to a local minimum, as the regrets barely decrease. The average results obtained by CBO, SCBO, and FDBO within the same maximum iterations are 12.9%, 10.5%, and 1.0% higher than the one via ASBO-I, respectively.

The constraint functions of the NHAB problem (Case 3) are the most complex among Cases 1–3. Both SCBO and FDBO fail to obtain feasible solutions within 100 independent runs, as each requires feasible initial solutions to guide further searches. However, in Case 3, no

TABLE 1 Result summary for Case 1 (NNP), Case 2 (Newbranin), and Case 3 (NHAB).

Algorithm	Problem	Number of initial samples	Iteration number	Evaluation number (average/best/worst)	Failure number	Solution results (average/best/worst)
ASBO-I	NNP	50	50	100	0	$(3.30/0.0084/12.65) \times 10^{-6}$
CBO	NNP	50	50	100	0	$(1.49/0.0060/6.45) \times 10^{-6}$
SCBO	NNP	50	50	74/52/100 ^a	0	$(5.37/0.0048/30.75) \times 10^{-6}$
FDBO	NNP	50	50	65/62/79 ^a	0	$(1.97/0.062/17.21) \times 10^{-14}$
ASBO-I	new Branin	100	50	150	0	−268.64/−268.69/−268.49
CBO	new Branin	100	50	150	0	−234.13/−267.18/−181.88
SCBO	new Branin	100	50	109/102/133 ^a	0	−240.31/−268.02/−188.54
FDBO	new Branin	100	50	150	0	−265.96/−268.57/−245.75
ASBO-I	NHAB	100	200	300	16	5.55/−4.48/18.27
CBO	NHAB	100	200	300	51	4.88/−2.46/9.19
FDBO	NHAB	100	200	300	100	—
SCBO	NHAB	100	200	300	100	—

^aThe algorithm reaches other termination conditions before reaching the maximum iterations.

feasible solution is identified across the total 10,000 samples (100 initial samples $\times$ 100 runs). We also record the number of runs in which feasible solutions have been found at the i th iteration across 100 runs, as shown by dashed lines in Figure 2C. CBO find the first feasible solution at the 13th iteration in 2 out of 100 runs, while ASBO-I finds the first feasible solution at the 1st iteration in 3 out of 100 runs. Only 49 out of 100 runs by CBO find feasible solutions, due to the unreasonable weight of the impact of the constraints and the objective function. However, 84 out of 100 runs by ASBO-I find feasible solutions. While the average solution found by CBO is smaller than that of ASBO-I, the best solution found by ASBO-I is better than the one by CBO. More importantly, the number of failed runs by ASBO-I is much smaller for ASBO-I compared to CBO.

4.1.2 | Comparisons with GA, PSO, SLSQP, and trust-constr

Additionally, the performance is compared with two commonly used stochastic algorithms: genetic algorithm (GA)⁵⁰ and PSO algorithm,⁵¹ and two gradient-based optimizers: sequential least squares programming (SLSQP)⁵² and trust region algorithm (Trust-constr).^{53,54} For PSO and GA, no initial points are required, and the maximum iterations are the same as the ones of BO algorithms. Note that the number of black-box function evaluations for PSO and GA is generally much larger than the maximum iterations, as it depends on other hyperparameters (e.g., swarm size). The hyperparameters of PSO and GA are detailed in <https://github.com/NicolasHong/ASBO>. For deterministic algorithms (SLSQP and Trust-constr), results depend on the initial values, which are generated randomly. Similarly, the algorithms are carried out 100 times using different initial values. Figures 3A–C show the boxplot of Cases 1 to 3 by all eight evaluated algorithms. The thick horizontal line within each box represents the median result (50th percentile) of the algorithm. The box spans the interquartile range (IQR), from the 25th percentile

(lower quartile) to the 75th percentile (upper quartile). The average iterations and the average solutions can be found in Table S1 of Section S4 of the Supporting Information.

For Case 1, all algorithms can obtain solutions close to the optimal in each run. In Case 2, all algorithms achieve the global optimal solution in 100 runs. However, the average results obtained by CBO, SCBO, FDBO, PSO, GA, SLSQP, and Trust-constr are 12.9%, 10.5%, 1.0%, 0.7%, 3.3%, 5.6%, and 11.4% higher than the ones obtained by ASBO-I, respectively. Additionally, the corresponding black-box evaluations for PSO and GA exceed those of BO algorithms by over tenfold.

In Case 3, PSO finds feasible solutions in 82 out of 100 runs, but the average solution is 200% larger than that of ASBO-I. Meanwhile, only 1 out of 100 runs by GA find a feasible solution. Similarly, the corresponding black-box evaluations of PSO and GA are significantly larger. Regarding deterministic algorithms, SLSQP and Trust-constr, only 2 and 22 runs fail out of 100 runs, respectively. The number of iterations required by SLSQP is smaller than that required by Trust-constr. However, due to their reliance on initial values, there is a considerable disparity between the best and worst solutions obtained, making the algorithms less stable. The average solutions obtained by SLSQP and Trust-constr are 95% and 126% higher than the ones by ASBO-I, respectively.

Overall, PSO and GA require numerous black-box evaluations to find feasible solutions. However, the high number of required evaluations makes them unsuitable for black-box models with high computational costs. As for SLSQP and Trust-constr, as shown in Figure 3, their results are heavily influenced by the initial values, with differences in the solutions obtained under different initial conditions.

4.1.3 | Hyperparameter sensitivity analysis

The influences of four key hyperparameters of Equation (10) (m_1, m_2, P_1 , and P_2) are investigated. Each hyperparameter is assigned

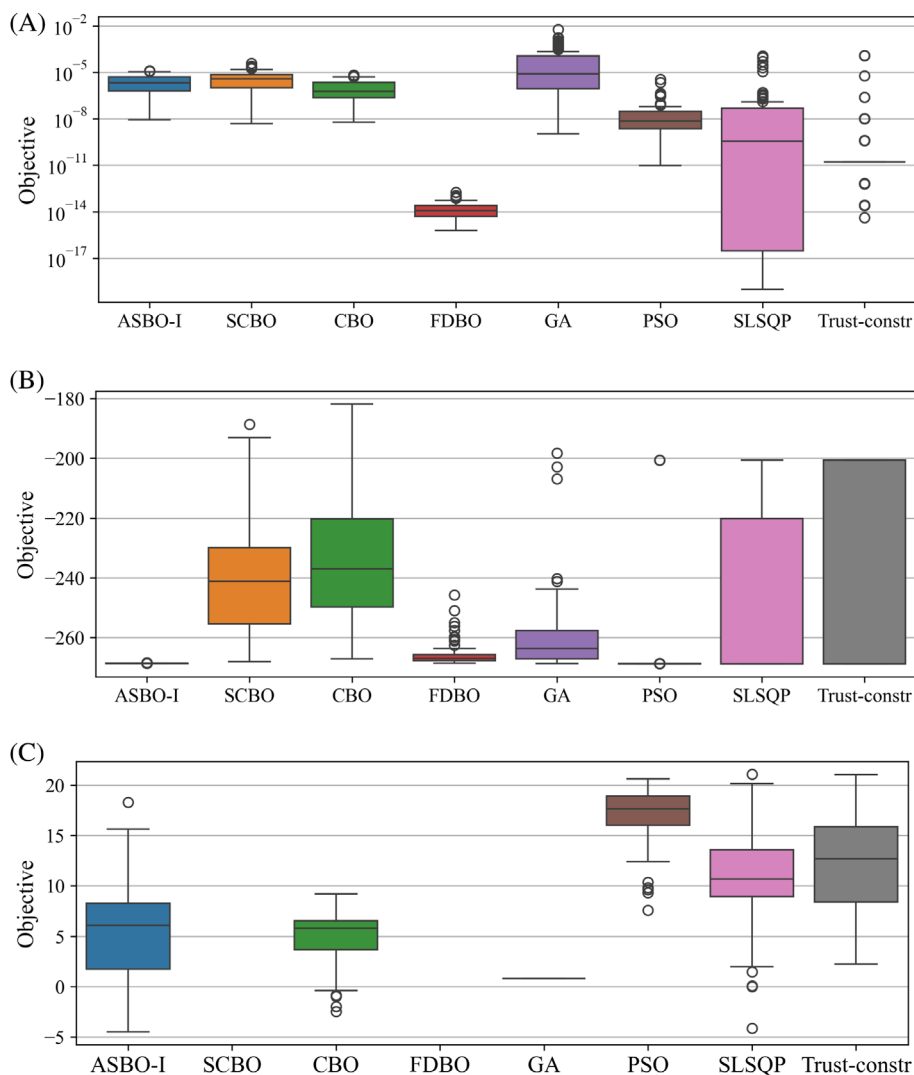


FIGURE 3 Boxplot of (A) Case 1, (B) Case 2, and (C) Case 3. The thick horizontal line within each box represents the median result (50th percentile) of the algorithm. The box spans the interquartile range (IQR), from the 25th percentile (lower quartile) to the 75th percentile (upper quartile). The whiskers extend from the edges of the box to the smallest and largest values within 1.5 times the IQR from the lower and upper quartiles. The points outside the whiskers represent outliers. There are no feasible solutions for Case 3 by SCBO and FDBO.

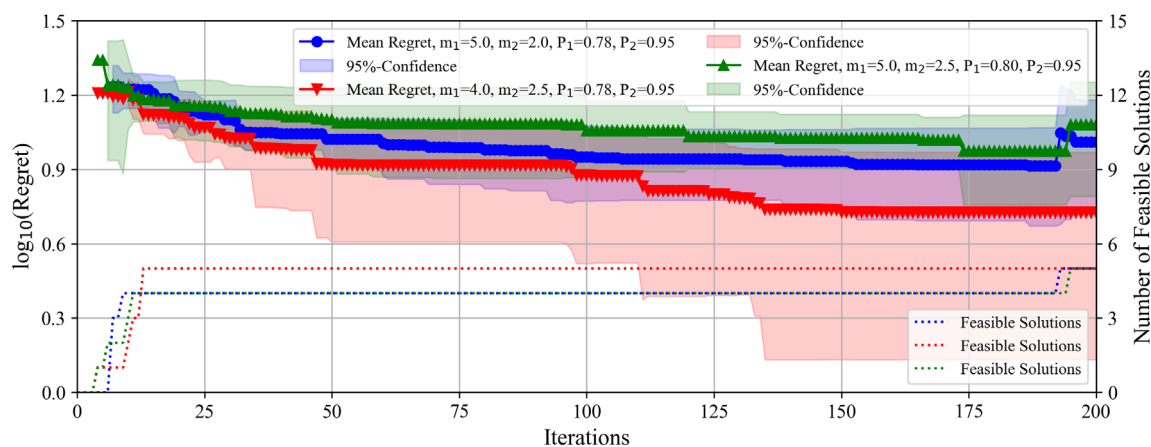


FIGURE 4 The $\log_{10}(\text{Regret})$ via ASBO-I when $m_1 = 5.0, m_2 = 2.0, P_1 = 0.78, P_2 = 0.95$; $m_1 = 4.0, m_2 = 2.5, P_1 = 0.78, P_2 = 0.95$; $m_1 = 5.0, m_2 = 2.5, P_1 = 0.80, P_2 = 0.95$.

three alternative values: $m_1 = 3.0, 4.0, 5.0, m_2 = 1.5, 2.0, 2.5, P_1 = 0.76, 0.78, 0.80, P_2 = 0.92, 0.95, 0.98$. Using Latin hypercube sampling, an additional 30 combinations of these values, excluding the

combination $m_1 = 5.0, m_2 = 2.0, P_1 = 0.78, P_2 = 0.95$, are generated. Hyperparameter combinations can be found in Section S5 of the Supporting Information. For each combination, ASBO-I is executed five

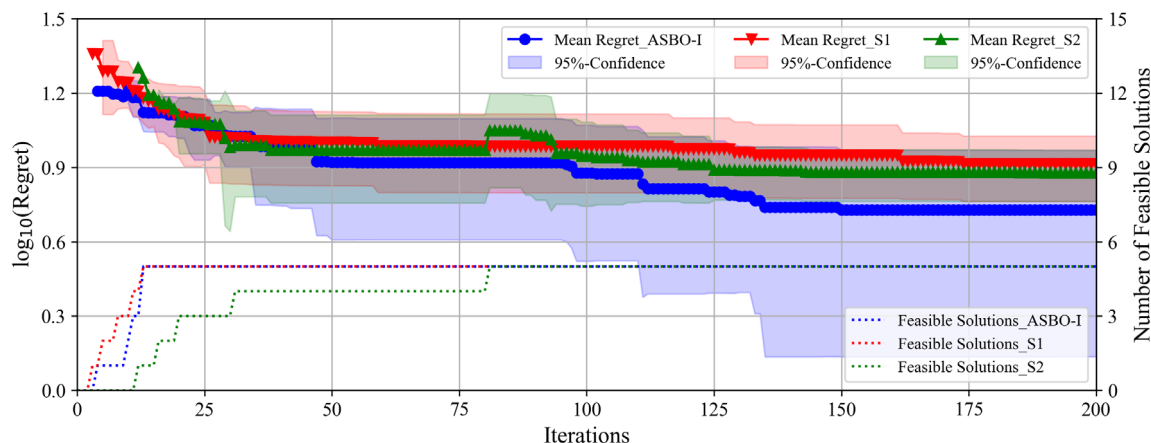


FIGURE 5 The $\log_{10}(\text{Regret})$ via ASBO-I, Strategy S1, and Strategy S2.

TABLE 2 Utility prices for EDP optimization problem.

Type	Cost	Inlet temperature ($^{\circ}\text{C}$)	Outlet temperature ($^{\circ}\text{C}$)
Cooling water	0.05 \$/ton	32	38
Low-pressure steam	39.7 \$/ton	180	179

times to estimate the average $\log_{10}\text{Regret}_i$ for Case 3. The number of initial points and iterations is set to 100 and 200, respectively. The regret plot and result summary for all 30 combinations are provided in Section S5 of the Supporting Information, with three representative cases displayed in Figure 4. These include the combinations with the best ($m_1 = 4.0, m_2 = 2.5, P_1 = 0.78, P_2 = 0.95$) and worst ($m_1 = 5.0, m_2 = 2.5, P_1 = 0.80, P_2 = 0.95$) average results, depicted in red and green, respectively, as well as the case using the hyperparameters from Equation (10), shown in blue. The increase in $\log_{10}\text{Regret}_i$ for the blue and green curves during the last few iterations is attributed to one of the five runs identifying a feasible solution higher than the other four runs.

The 30 combinations are grouped m_1 , m_2 , P_1 , and P_2 , respectively, with detailed results provided in Section S5 of the Supporting Information. When comparing $m_1 = 3.0$, $m_1 = 4.0$, and $m_1 = 5.0$, the smallest average $\log_{10}\text{Regret}_i$ is observed for $m_1 = 4.0$, followed by $m_1 = 3.0$. Similarly, for $P_1 = 0.76$, $P_1 = 0.78$, and $P_1 = 0.80$, the smallest average $\log_{10}\text{Regret}_i$ is achieved with $P_1 = 0.78$, followed by $P_1 = 0.80$. In contrast, the differences in results for $m_1 = 3.0, 4.0, 5.0$ and for $P_2 = 0.92, 0.95, 0.98$ are relatively small. The sensitivity analysis reveals that m_1 and P_1 have a more significant impact on the algorithm performance.

4.1.4 | Comparisons of three-stage adaptive sampling and single-stage sampling

To demonstrate the advantage of the proposed three-stage adaptive acquisition function, two cases are carried out using only the acquisition functions in the first and second stages of Equation (10). Figure 5 illustrates the $\log_{10}(\text{Regret})$ for the acquisition functions defined in Equations (12) and (13), referred to strategies S1 and S2, respectively.

The performance of ASBO-I using Equation (10) with $m_1 = 4$ and $m_2 = 2.5$ is also shown in Figure 5. ASBO-I, strategy S1, and strategy S2 are denoted by blue, red, and green, respectively. When using single-stage acquisition functions (S1 and S2), the algorithms are more prone to convergence to local minima, as indicated by the smaller gradients of the red and green curves. The number of runs in which feasible solutions have been found at the i th iteration across five runs is also shown in Figure 5 by dashed lines. A key difference between S1 and S2 lies in their feasibility exploration capabilities: Strategy S1 demonstrates a stronger ability to explore feasibility, identifying the first feasible solution earlier, whereas strategy S2 does not find one until the 12th iteration:

$$AC = \text{UCB}(\mu, \sigma, \kappa) + \frac{1}{s + TP_{\text{constraint}}^m} \quad m = 4, s = 10^{-5}, \kappa = 2.5 \quad (12)$$

$$AC = \text{UCB}(\mu, \sigma, \kappa) + \frac{1}{s + TP_{\text{constraint}}^m} \quad m = 2.5, s = 10^{-5}, \kappa = 2.5 \quad (13)$$

4.2 | Case 4 (extractive distillation process optimization, EDP)

In Case 4, we address an extractive distillation process optimization problem, adapted from Ma et al.⁴⁶ The process comprises two distillation columns (C1 and C2), where the extractive solvent phenol is utilized to help separate the azeotropic mixture of toluene and n-heptane. Detailed process introduction can be found in Ma et al.⁴⁶ The objective is to minimize the energy cost of two distillation columns. The utility

prices used for the condenser and reboiler of the distillation tower are shown in Table 2, while the lower and upper bounds for the decision variables and the constraints are shown in Table 3. The decision variables include the mole flow of phenol in the solvent stream (F_s), reflux ratio of column C1 (r_1), distillate rate of column C1 (D_1), reflux ratio of column C2 (r_2), and distillate rate of column C2 (D_2). The constraints are the mole fraction of n-heptane in the distillate stream of column C1 (x_n) and the mole fraction of toluene in the distillate stream of column C2 (x_t). The process is modeled using Aspen Plus. For the given decision variable bounds, the convergence of the extractive distillation process in Aspen Plus cannot be guaranteed, since the convergence of distillation columns is highly sensitive to these variables. It significantly impacts the stability of the algorithm. To address this issue and ensure stable convergence of the simulator, an infeasible path method⁵⁵ is adopted.

According to the analysis results for Cases 1–3, the ASBO-I and SLSQP algorithms outperform the other algorithms. Thus, only the ASBO-I and SLSQP algorithms are used to solve Case 4. Additionally, the SQP and BOBYQA⁵⁶ (Bound Optimization BY Quadratic Approximation) optimizers inside Aspen Plus are used for comparisons.

ASBO-I, BOBYQA, SQP, and SLSQP are carried out 10 times with different initial values. The ASBO-I algorithm uses 100 random initial

values and undergoes 60 iterations in each run, while the SLSQP, SQP, and BOBYQA algorithms use the same initial value in each run. The solution results and the corresponding iteration numbers are shown in Table 4. Due to the susceptibility of the convergence process to decision variables, methods relying on numerical differentiation are prone to failure. Notably, while SLSQP performs well in benchmark problems, it does not work well with numerical accuracy issues in simulation models. SLSQP and SQP in Aspen fail to solve the problem 10 and 7 out of 10 runs, respectively. In contrast, ASBO-I and BOBYQA within Aspen achieve feasible solutions in all ten runs. The best solution obtained by BOBYQA is about 8% lower than the one by ASBO-I; however, BOBYQA performance highly relies on initial points. Its worst solution is more than twice as high as its best solution. On the other hand, the ASBO-I algorithm demonstrates robust performance, with the worst solution being only 21% higher than the best solution.

4.3 | Case 5 (solution polyethylene process optimization, SPE)

In Case 5, an industrial solution polyethylene process optimization problem constructed in Aspen Plus is adopted. The process flow diagram of the process is shown in Figure 6. The reactor is an adiabatic reactor whose temperature is controlled by adjusting the cold feed temperature. The reactor temperature should be maintained at a temperature for high catalyst activity. The process involves intricate internal recycling streams, polymerization kinetics, and thermodynamic

TABLE 3 Upper and lower bounds of decision variables and constraints in Case 4.

	F_s	r_1	D_1	r_2	D_2	x_n	x_t
Lower bound	40	3	30	0.1	20	0.98	0.98
Upper bound	100	10	80	5	80	1	1

TABLE 4 Calculation results for the EDP problem.

Algorithm	Problem	Number of initial samples	Iteration number (average/best/worst)	Failure number	Solution results (average/best/worst)
ASBO-I	EDP	100	60	0	(268.1/248.6/299.6) \$/h
SLSQP	EDP	—	—	10	—
BOBYQA (Aspen)	EDP	—	—	0	(303.3/228.8/504.1) \$/h
SQP (Aspen)	EDP	—	—	7	(229.9/229.4/230.8) \$/h [#]

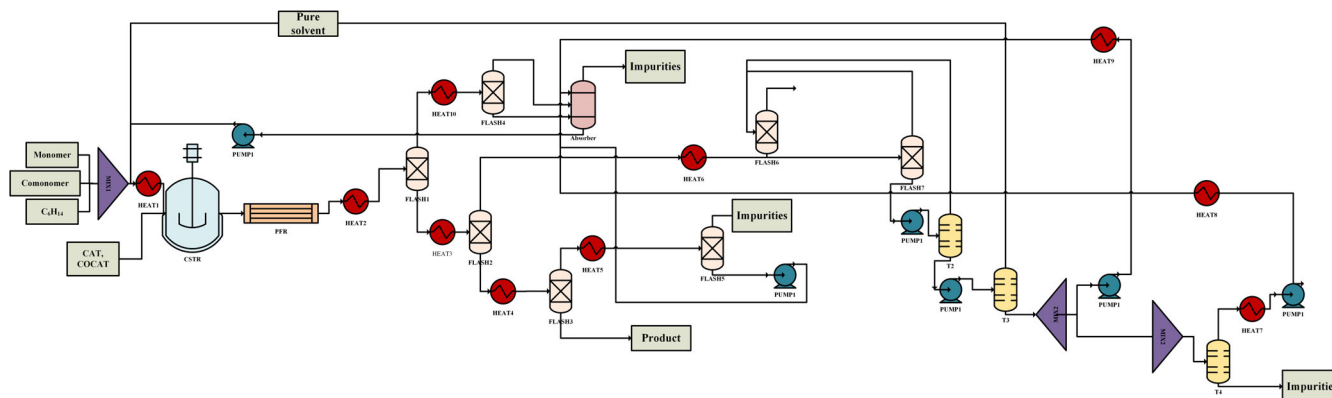


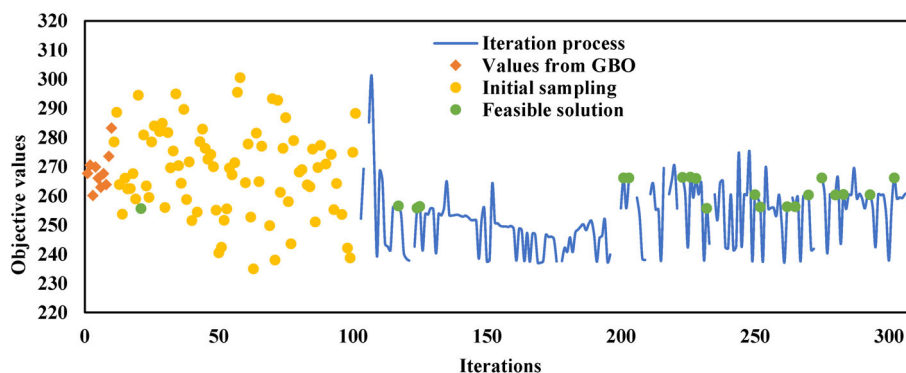
FIGURE 6 The process flow diagram of the solution polyethylene process (SPE).

TABLE 5 Problem statement of the SPE problem.

Decision variables	Constraint	Object function
Ethylene flow	T4 pressure	Utility prices
Octene flow	T3 reflux ratio	+Raw material prices
Hexane flow rate	T1 pressure	
Tear stream flow	FLASH1 pressure	
Reactor inlet temperature	FLASH2 pressure	

TABLE 6 Results of various algorithms for the SPE problem.

Algorithm	Problem	Initial points	Evaluation number	Result
SLSQP	SPE	—	—	—
ASBO-I	SPE	100	200	—
ASBO-II	SPE	100 + 10	200	255.62 \$/h

FIGURE 7 Schematic diagram of the iteration process of the ASBO-II algorithm.

calculations, rendering it a nonlinear black-box model. The ASBO-I, ASBO-II, and SLSQP algorithms are used to solve Case 5. The problem statement is shown in Table 5, including 10 decision variables and five constraints. The decision variables include feed components, flash pressure, column pressure of the distillation tower, and reflux ratio. The constraints include process requirements, capacity requirements, and the convergence of the tearing stream. The objective function is the sum of the utility prices and the unit price of raw materials. Utility prices and the purity and price of raw materials are provided in Section S6 of the Supporting Information.

As shown in Table 6, the SLSQP algorithm is used to solve this problem. Due to the high computational cost of single process simulation, only 10 runs are carried out by the SLSQP algorithm using different initial variables. All 10 runs fail to converge within the maximum iteration number of 500. The ASBO-I algorithm fails to find feasible solutions within 10 runs using 100 initial points and 200 iterations. Then, the ASBO-II algorithm is applied. A gradient-based optimizer (SLSQP) is first used to solve the feasibility problem using 10 different initial values, and the maximum iteration number of each feasibility problem is 300. The obtained solutions are then input into BO as initial samples, and the lower and upper bounds of decision variables are readjusted based on the results. In this complex case, m_1 of Equation (10) is adjusted to 6 to prioritize the search for feasible solutions. The iteration process in Figure 7 shows a feasible solution (green dot)

is found from the 100 initial samples (orange dot) after the decision variables are readjusted. In the following 200 iterations (blue line), 20 feasible solutions (green dots) are obtained.

5 | CONCLUSION

This article proposes an adaptive sampling Bayesian optimization (ASBO) algorithm for CBBO problems, demonstrating stable convergence. A three-stage adaptive acquisition function is proposed to explore feasible solutions, balance feasibility and optimality, and optimize the objective. A hybrid algorithm, ASBO-II, is developed for complex CBBO, utilizing a gradient-based optimizer (GBO) and the Bayesian optimization (ASBO-I). Solving a feasibility problem using a gradient-based optimizer can target the feasibility domain efficiently and construct an accurate posterior distribution, addressing the cold-start problem commonly encountered in BO algorithms.

The effectiveness of the proposed algorithm is validated through five constrained problems, with comparisons made against commonly used stochastic algorithms, deterministic algorithms, and advanced BO algorithms from the literature. In three benchmark problems (Cases 1–3), ASBO-I achieves global or local optimal solutions with less iterations. In two process optimization problems (Case 4 and Case 5), ASBO-I shows better stability than two Aspen internal algorithms,

BOBYQA and SQP. Furthermore, the hybrid algorithm of ASBO-II effectively addresses an extremely complex CBBO problem. The CBBO problems, where feasible solutions cannot be easily obtained by initial sampling, can truly benefit from the proposed hybrid algorithm. The case studies clearly show that the ASBO methods exhibit better convergence performances and find better solutions.

In constrained optimization, it is common for one particularly complex constraint to be unsatisfied while all others are met. Our current acquisition function is influenced by all constraints equally, making it challenging to pinpoint the critical constraint driving feasibility. Our future work will focus on developing an adaptive sampling strategy that can prioritize sampling based on key constraints.

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DATA AVAILABILITY STATEMENT

The numerical data from Figures 2–5, and 7 are tabulated in the Supplementary Material.

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REFERENCES

1. Caballero JA, Grossmann IE. An algorithm for the use of surrogate models in modular flowsheet optimization. *AIChE J.* 2008;54(10):2633–2650. doi:[10.1002/aic.11579](https://doi.org/10.1002/aic.11579)
2. Rios LM, Sahinidis NV. Derivative-free optimization: a review of algorithms and comparison of software implementations. *J Glob Optim.* 2013;56(3):1247–1293. doi:[10.1007/s10898-012-9951-y](https://doi.org/10.1007/s10898-012-9951-y)
3. Schweidtmann AM, Mitsos A. Deterministic global optimization with artificial neural networks embedded. *J Optim Theory Appl.* 2019;180(3):925–948. doi:[10.1007/s10957-018-1396-0](https://doi.org/10.1007/s10957-018-1396-0)
4. Eason JP, Biegler LT. Reduced model trust region methods for embedding complex simulations in optimization problems. In: Gernaey KV, Huusom JK, Gani R, eds. *Computer Aided Chemical Engineering*. Vol 37. 12. Elsevier; 2015:773–778. doi:[10.1016/B978-0-444-63578-5.50124-9](https://doi.org/10.1016/B978-0-444-63578-5.50124-9)
5. Agarwal A, Biegler LT. A trust-region framework for constrained optimization using reduced order modeling. *Optim Eng.* 2013;14(1):3–35. doi:[10.1007/s11081-011-9164-0](https://doi.org/10.1007/s11081-011-9164-0)
6. Schmit LA, Farshi B. Some approximation concepts for structural synthesis. *AIAA J.* 1974;12(5):692–699. doi:[10.2514/3.49321](https://doi.org/10.2514/3.49321)
7. Wang H, Jin Y, Sun C, Doherty J. Offline data-driven evolutionary optimization using selective surrogate ensembles. *IEEE Trans Evol Comput.* 2019;23(2):203–216. doi:[10.1109/TEVC.2018.2834881](https://doi.org/10.1109/TEVC.2018.2834881)
8. Christopher CCE, Dutta A, Farooq S, Karimi IA. Process synthesis and optimization of propylene/propane separation using vapor recompression and self-heat recuperation. *Ind Eng Chem Res.* 2017;56(49):14557–14564. doi:[10.1021/acs.iecr.7b03432](https://doi.org/10.1021/acs.iecr.7b03432)
9. Javaloyes-Antón J, Kronqvist J, Caballero JA. Simulation-based optimization of distillation processes using an extended cutting plane algorithm. *Comput Chem Eng.* 2022;159:107655. doi:[10.1016/j.compchemeng.2021.107655](https://doi.org/10.1016/j.compchemeng.2021.107655)
10. Kleijnen JPC. Design and analysis of simulation experiments. In: Pilz J, Rasch D, Melas VB, Moder K, eds. *Statistics and Simulation*. Springer International Publishing; 2018:3–22. doi:[10.1007/978-3-319-76035-3_1](https://doi.org/10.1007/978-3-319-76035-3_1)
11. Jiang P, Zhou Q, Shao X. Surrogate-model-based design and optimization. In: Jiang P, Zhou Q, Shao X, eds. *Surrogate Model-Based Engineering Design and Optimization*. Springer; 2020:135–236. doi:[10.1007/978-981-15-0731-1_7](https://doi.org/10.1007/978-981-15-0731-1_7)
12. Cozad A, Sahinidis NV, Miller DC. Learning surrogate models for simulation-based optimization. *AIChE J.* 2014;60(6):2211–2227. doi:[10.1002/aic.14418](https://doi.org/10.1002/aic.14418)
13. Frazier PI. A tutorial on Bayesian optimization. Published Online July 8 2018. Accessed September 18, 2023 <http://arxiv.org/abs/1807.02811>
14. Shahriari B, Swersky K, Wang Z, Adams RP, de Freitas N. Taking the human out of the loop: a review of Bayesian optimization. *Proc IEEE.* 2016;104(1):148–175. doi:[10.1109/JPROC.2015.2494218](https://doi.org/10.1109/JPROC.2015.2494218)
15. Wang Z, Jegelka S. Max-value entropy search for efficient Bayesian optimization. *Proceedings of the 34th International Conference on Machine Learning*. PMLR; 2017:3627–3635 Accessed July 13, 2024. <https://proceedings.mlr.press/v70/wang17e.html>
16. Bartoli N, Lefebvre T, Dubreuil S, et al. Adaptive modeling strategy for constrained global optimization with application to aerodynamic wing design. *Aerosp Sci Technol.* 2019;90:85–102. doi:[10.1016/j.ast.2019.03.041](https://doi.org/10.1016/j.ast.2019.03.041)
17. Urm JJ, Choi JH, Kim C, Lee JM. Techno-economic analysis and process optimization of a PET chemical recycling process based on Bayesian optimization. *Comput Chem Eng.* 2023;179:108451. doi:[10.1016/j.compchemeng.2023.108451](https://doi.org/10.1016/j.compchemeng.2023.108451)
18. Ward A, Pini R. Efficient Bayesian optimization of industrial-scale pressure-vacuum swing adsorption processes for CO₂ capture. *Ind Eng Chem Res.* 2022;61(36):13650–13668. doi:[10.1021/acs.iecr.2c02313](https://doi.org/10.1021/acs.iecr.2c02313)
19. Tian H, Ierapetritou MG. A surrogate-based framework for feasibility-driven optimization of expensive simulations. *AIChE J.* 2024;70(5):e18364. doi:[10.1002/aic.18364](https://doi.org/10.1002/aic.18364)
20. Durantin C, Marzat J, Balesdent M. Analysis of multi-objective kriging-based methods for constrained global optimization. *Comput Optim Appl.* 2016;63(3):903–926. doi:[10.1007/s10589-015-9789-6](https://doi.org/10.1007/s10589-015-9789-6)
21. Parr JM, Keane AJ, Forrester AIJ, Holden CME. Infill sampling criteria for surrogate-based optimization with constraint handling. *Eng Optimiz.* 2012;44(10):1147–1166. doi:[10.1080/0305215X.2011.637556](https://doi.org/10.1080/0305215X.2011.637556)
22. Priem R, Bartoli N, Diouane Y. On the use of upper trust bounds in constrained Bayesian optimization infill criteria. *AIAA Aviation 2019 Forum*. American Institute of Aeronautics and Astronautics; 2019. doi:[10.2514/6.2019-2986](https://doi.org/10.2514/6.2019-2986)
23. Paulson JA, Lu C. COBAL: COstrained Bayesian optimizAtion of computationally expensive grey-box models exploiting derivative information. *Comput Chem Eng.* 2022;160:107700. doi:[10.1016/j.compchemeng.2022.107700](https://doi.org/10.1016/j.compchemeng.2022.107700)
24. Lu C, Paulson JA. No-regret Bayesian optimization with unknown equality and inequality constraints using exact penalty functions. *IFAC-PapersOnLine.* 2022;55(7):895–902. doi:[10.1016/j.ifacol.2022.07.558](https://doi.org/10.1016/j.ifacol.2022.07.558)
25. Jiao R, Zeng S, Li C, Jiang Y, Jin Y. A complete expected improvement criterion for Gaussian process assisted highly constrained expensive optimization. *Inform Sci.* 2019;471:80–96. doi:[10.1016/j.ins.2018.09.003](https://doi.org/10.1016/j.ins.2018.09.003)
26. Li Y, Wu Y, Zhao J, Chen L. A kriging-based constrained global optimization algorithm for expensive black-box functions with infeasible initial points. *J Glob Optim.* 2017;67(1):343–366. doi:[10.1007/s10898-016-0455-z](https://doi.org/10.1007/s10898-016-0455-z)
27. Kudva A, Sorourifar F, Paulson JA. Constrained robust Bayesian optimization of expensive noisy black-box functions with guaranteed regret bounds. *AIChE J.* 2022;68(12):e17857. doi:[10.1002/aic.17857](https://doi.org/10.1002/aic.17857)
28. Bajaj I, Iyer SS, Faruque Hasan MM. A trust region-based two phase algorithm for constrained black-box and grey-box optimization with infeasible initial point. *Comput Chem Eng.* 2018;116:306–321. doi:[10.1016/j.compchemeng.2017.12.011](https://doi.org/10.1016/j.compchemeng.2017.12.011)

29. Lam R, Willcox K. Lookahead Bayesian optimization with inequality constraints. *Advances in Neural Information Processing Systems*. Vol 30. Curran Associates, Inc.; 2017. Accessed July 13, 2024. <https://proceedings.neurips.cc/paper/2017/hash/83f97f4825290be4cb794ec6a234595f-Abstract.html>
30. Lam R, Allaire DL, Willcox KE. Multifidelity optimization using statistical surrogate modeling for non-hierarchical information sources. In: *56th AIAA/ASCE/AHS/ASC Structures, Structural Dynamics, and Materials Conference*. 0143. 2015.
31. Rasmussen CE, Williams CKI. *Gaussian Processes for Machine Learning*. MIT Press; 2006.
32. Liu J, Han Z, Song W. Comparison of infill sampling criteria in kriging-based aerodynamic optimization. *28th Congress of the International Council of the Aeronautical Sciences*. Vol 2; 2012:1625-1634. https://icas.org/icas_archive/ICAS2012/PAPERS/269.PDF
33. Srinivas N, Krause A, Kakade SM, Seeger M. Gaussian process optimization in the bandit setting: No regret and experimental design. *IEEE Trans Inform Theory*. 2012;58(5):3250-3265. doi:10.1109/TIT.2011.2182033
34. Jones DR, Schonlau M. Efficient global optimization of expensive black-box functions. *J Global Optimiz*. 1998;13:455-492.
35. Kushner HJ. A new method of locating the maximum point of an arbitrary multipiece curve in the presence of noise. *J Basic Eng*. 1964; 86(1):97-106. doi:10.1115/1.3653121
36. Dong H, Song B, Wang P, Huang S. A kind of balance between exploitation and exploration on kriging for global optimization of expensive functions. *J Mech Sci Technol*. 2015;29(5):2121-2133. doi:10.1007/s12206-015-0434-1
37. Schonlau M, Welch WJ, Jones DR. Global versus local search in constrained optimization of computer models. *Lect Notes Ser*. 1998;34:11-25.
38. Bagheri S, Konen W, Allmendinger R, et al. Constraint handling in efficient global optimization. In: *Proceedings of the Genetic and Evolutionary Computation Conference*. GECCO '17. Association for Computing Machinery. 673-680. 2017. doi:10.1145/3071178.3071278
39. Sasena MJ. *Flexibility and Efficiency Enhancements for Constrained Global Design Optimization with Kriging Approximations*. University of Michigan; 2002.
40. Gelbart MA, Snoek J, Adams RP. Bayesian optimization with unknown constraints. Published Online March 21 2014. doi:10.48550/arXiv.1403.5607
41. Priem R, Bartoli N, Diouane Y, Sgueglia A. Upper trust bound feasibility criterion for mixed constrained Bayesian optimization with application to aircraft design. *Aerosp Sci Technol*. 2020;105:105980. doi:10.1016/j.ast.2020.105980
42. Ma Y, McLaughlan M, Zhang N, Li J. Novel feasible path optimisation algorithms using steady-state and/or pseudo-transient simulations. *Comput Chem Eng*. 2020;143:107058. doi:10.1016/j.compchemeng.2020.107058
43. Adams II TA. *Learn Aspen Plus in 24 Hours*. 2nd ed. McGraw-Hill Education; 2022 Accessed September 26, 2023. <https://www.accessengineeringlibrary.com/content/book/9781264266654>
44. Qian J, Cheng Y, Zhang J, Liu J, Zhan D. A parallel constrained efficient global optimization algorithm for expensive constrained optimization problems. *Eng Optimiz*. 2021;53(2):300-320. doi:10.1080/0305215X.2020.1722118
45. Garrido-Merchán EC, Fernández-Sánchez D, Hernández-Lobato D. Parallel predictive entropy search for multi-objective Bayesian optimization with constraints applied to the tuning of machine learning algorithms. *Expert Syst Appl*. 2023;215:119328. doi:10.1016/j.eswa.2022.119328
46. Ma K, Sahinidis NV, Bindlish R, Bury SJ, Haghpahan R, Rajagopalan S. Data-driven strategies for extractive distillation unit optimization. *Comput Chem Eng*. 2022;167:107970. doi:10.1016/j.compchemeng.2022.107970
47. Rogers A, Ierapetritou M. Feasibility and flexibility analysis of black-box processes part 1: surrogate-based feasibility analysis. *Chem Eng Sci*. 2015;137:986-1004. doi:10.1016/j.ces.2015.06.014
48. Gardner JR, Kusner MJ, Xu ZE, Weinberger KQ, Cunningham JP. Bayesian Optimization with Inequality Constraints. In: *31st International Conference on Machine Learning: ICML 2014, Beijing, China, 21-26 June 2014, Volume 3 of 5*. 2581-2591. 2014.
49. Eriksson D, Pearce M, Gardner J, Turner RD, Poloczek M. Scalable global optimization via local Bayesian optimization. *Advances in Neural Information Processing Systems*. Vol 32. Curran Associates, Inc; 2019. Accessed November 16, 2023. <https://proceedings.neurips.cc/paper/2019/hash/6c990b7aca7bc7058f5e98ea909e924b-Abstract.html>
50. Srinivas M, Patnaik LM. Adaptive probabilities of crossover and mutation in genetic algorithms. *IEEE Trans Syst Man Cybern*. 1994;24(4): 656-667. doi:10.1109/21.286385
51. Zhan ZH, Zhang J, Li Y, Chung HSH. Adaptive particle swarm optimization. *IEEE Trans Syst Man Cybern B Cybern*. 2009;39(6):1362-1381. doi:10.1109/TSMCB.2009.2015956
52. Gupta M, Gupta B. An Ensemble Model for Breast Cancer Prediction Using Sequential Least Squares Programming Method (SLSQP). In: *2018 Eleventh International Conference on Contemporary Computing (IC3)*. 1-3. 2018. doi:10.1109/IC3.2018.8530572
53. Byrd RH, Hribar ME, Nocedal J. An interior point algorithm for large-scale nonlinear programming. *SIAM J Optim*. 1999;9(4):877-900. doi:10.1137/S1052623497325107
54. Byrd RH, Gilbert JC, Nocedal J. A trust region method based on interior point techniques for nonlinear programming. *Math Program*. 2000;89(1):149-185. doi:10.1007/PL00011391
55. Biegler LT, Hughes RR. Infeasible path optimization with sequential modular simulators. *AIChE J*. 1982;28(6):994-1002. doi:10.1002/aic.690280615
56. Powell MJD. The BOBYQA algorithm for bound constrained optimization without derivatives. *Cambridge NA Report NA2009/06*. Vol 26. University of Cambridge; 2009:26-46.

SUPPORTING INFORMATION

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